ثانوية المدينة التعليمية
Education City High School

2.10

Inverses of Exponential Functions

Student Notes and Guided Practice

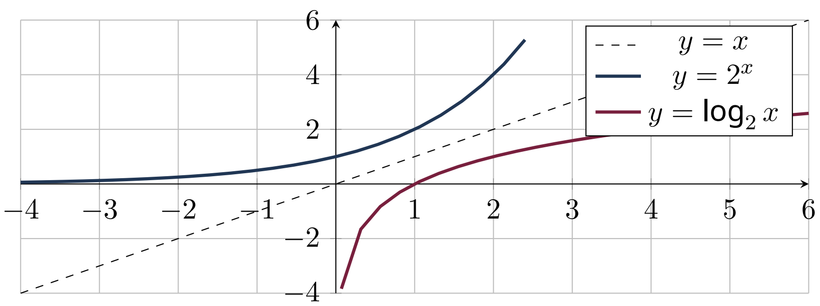
AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Explain logarithms as inverses of exponential functions.
- Connect inverse pairs through graphs, compositions, domains, and ranges.
- Find inverses of transformed exponential functions.

Essential Question

How does reversing an exponential relationship create a logarithmic function?



Concept Framework

Key Idea

Exponential domain and range swap to become logarithmic range and domain.

Key Idea

The horizontal exponential asymptote becomes a vertical logarithmic asymptote after reflection.

Key Idea

To find an inverse, isolate the exponential expression before taking a logarithm.

Key Idea

Composition verifies the inverse on appropriate domains.

Formula Map

Formula and Structure

$$y = ab^{x-h} + k \Rightarrow x = h + \log_b\left(\frac{y-k}{a}\right)$$

Formula and Structure

$$f^{-1}(x) = h + \log_b\left(\frac{x-k}{a}\right)$$

Worked Examples

Worked Example 1

Problem. Find the inverse of $f(x) = 2^{x-3} + 5$.
Solution. $f^{-1}(x) = 3 + \log_2(x-5)$, domain $x > 5$.

Worked Example 2

Problem. Identify the vertical asymptote of the inverse above.
Solution. $x = 5$, the reflection of the original horizontal asymptote $y = 5$.

Worked Example 3

Problem. Verify $\log_3(3^x) = x$.
Solution. By inverse-function composition, the log asks for the exponent on base 3 that produces 3^x, namely x.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Find the inverse of $f(x) = 2^{x-1} + 3$, including its domain.

2. Find the inverse of $f(x) = 5 \cdot 3^x - 2$, including its domain.

3. Find the inverse of $f(x) = -4(1/2)^x + 7$, including its domain.

4. Find the inverse of $f(x) = e^{2x+1}$, including its domain.

5. Find the inverse of $f(x) = 10^{x/3}$, including its domain.

6. Original exponential has horizontal asymptote y=5. What asymptote does its inverse have?

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is ______.
2. A restriction I must remember is ______.
3. One graphical feature I can justify algebraically is ______.

Video support: Flipped Math topic page | Original ECHS material informed by Pearson and Holt Precalculus concepts.